

Show all of your work in the space provided. Full credit will not be awarded for correct answers unless sufficient supporting work is presented.

1. (2 pts) Give a quick definition of what it means to say a vector space, V , is finite dimensional.

2.(12 pts) Let A be a 12×10 matrix with three **non-pivot** columns. What are the following values?

a) Rank $A =$

b) Nullity $A =$

c) Dimension of $\text{Col}(A) =$

d) Dimension of $\text{Col}(A^T) =$

e) Dimension of $\text{Row}(A) =$

f) Dimension of $\text{Nul}(A^T) =$

3.(10 pts) Let $A = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$.

a) What is the characteristic equation for A ?

b) What are the eigenvalues for A ?

4. (10 pts) A scientist finds a solution to a nonhomogeneous system $A\mathbf{x} = \mathbf{b}$ of 10 equations with 8 unknowns. Using this fact, **decide if each statement below is impossible, possible but not guaranteed, or must be true.** Circle the correct choice and give a very brief explanation of your answer.

a) The system has a solution for all choices of $\mathbf{b}$.

impossible

possible but not guaranteed

must be true

Explain:

b) For this particular choice of $\mathbf{b}$ the system has infinitely many solutions.

impossible

possible but not guaranteed

must be true

Explain:

5.(16 pts) Let $A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{bmatrix}$. The eigenvalues of A are 1 and 5.

Is A diagonalizable? If so, find a matrix P such that $A = PDP^{-1}$.

6.(12 pts) Let A be a 3×3 matrix with eigenvalues 1, -2, 3, with corresponding

eigenvectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} -3 \\ -3 \\ 0 \end{bmatrix}$. Let $\mathbf{x} = \begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix}$

a) Find the coordinate vector for $\mathbf{x}$ relative to the basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

b) Find the value of $A^3\mathbf{x}$.

7. (8 pts) If $\mathbf{x}$ is an eigenvector for A with eigenvalue λ , show that $\mathbf{x}$ is also an eigenvector for $5A$.

8. (4 pts) Let W be a subspace of $\mathbb{R}^n$. What does it mean for the vector $\mathbf{x}$ to be in the orthogonal complement of W ?

9. (10 pts) Let $\mathbf{u} = \begin{bmatrix} 6 \\ 4 \\ -3 \end{bmatrix}$.

a) Give an example of a nonzero vector that is orthogonal to $\mathbf{u}$.

b) Find a unit vector in the opposite direction of $\mathbf{u}$.

10. Let W be the subspace spanned by the orthogonal vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$

a) (6 pts) What is the projection of the vector $\mathbf{y} = \begin{bmatrix} 3 \\ -2 \\ 4 \\ -3 \end{bmatrix}$ onto W ?

b) (4 pts) Write $\mathbf{y}$ as the sum of a vector in W and a vector in $W^\perp$.

11. (6 pts) Let A be any $m \times n$ matrix. The four fundamental subspaces associated with matrix A are listed in the column on the left. Several other subspaces associated with A are listed in the column on the right. Draw a line between subspaces that are equal.

Row(A)

$(\text{Row}(A))^{\perp}$

$(\text{Col}(A))^{\perp}$

Col(A)

$(\text{Null}(A))^{\perp}$

Null(A)

$(\text{Null}(A^T))^{\perp}$

Row(A^T)

Null(A^T)

Col(A^T)